



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df002-2017-android-2
Operator:	Floris
Analysis Started:	2026-06-19 10:52:51
Analysis Completed:	2026-06-19 10:53:53
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df002\2017\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df002\2017\dfdof_output\ft-df002-2017-android-2
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df002_external.001	drone_sd	physical	7.4 GB	2026-06-19 10:53:19
df002_flight.zip	controller_android	logical	134.1 MB	2026-06-19 10:53:20

Evidence Source Path

C:\Users\Floris\Documents\ddj_drones\train_phantom_3\df002\2017\android

Cryptographic Hashes

df002_external.001		drone_sd	physical
SHA-256:	c524d08c7bcb4d1dde6a20ff53e11c8f68534be45c9dfa8f4265e802ace104d1		
SHA-1:	8588c2882c6afad1b61b2d98d1a619c00ce9d160		

df002_flight.zip		controller_android	logical
SHA-256:	b991fe13218b2b82f193c932365523815fc52e961d195304c35cdf740b632a99		
SHA-1:	b580b478b2665d5d1ee2631f2e82bc64f937cf28		

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for df002_flight.zip
2. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for df002_flight.zip
3. [p2 - controller ios]: source evidence not found
4. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller android - images]: no artefacts found
7. [p3 - controller ios]: source evidence not found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	6	4	[~] Partial
fls	1	1	[OK]
icat	5	5	[OK]
mmls	1	1	[OK]
parser_android	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	6 / 7
Flights With Primary Correlation	0 / 1
Tools Succeeded	5 / 6

3. Device and Account Identification

Field	Value	Confidence	Sources
Drone Serial Number	03Z1419944	[K] Controller-only	2
Drone Model	P3 Standard	[K] Controller-only	1
DJI Application Version	3.1.6	[K] Controller-only	1
Product Type	2	[K] Controller-only	1
Firmware Version	1.3.50+	[K] Controller-only	1
Account UID	879825329191059456	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 26e38258c0d1b7cac499d46595880a3d6ac2699aa417c56cbb8700836e6934fc
Flight Record Name: DJIFlightRecord_2017-06-29_[13-39-52].csv
Start: 2017-06-29T19:39:52.161000+00:00
End: 2017-06-29T19:49:43.057000+00:00
Duration: 590.9 s

Flight Metrics

Peak Height: 32.6 m
Distance Travelled: 1399.61 m
Video Recording: Yes
Photo Capture: Yes
Photos Taken: 1
Recording Duration: 0.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. GPS_Atti
2. AutoTakeoff
3. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-06-29 19:39:52	ctrl_and:fr (v3.1.6)	Log started	SN: 03Z1419944; App: 3.1.6; P3 Standard
2017-06-29 19:39:52	ctrl_and:fr (v3.1.6)	Motor turned on	on
2017-06-29 19:39:55	ctrl_and:fr (v3.1.6)	Record mode changed	No
2017-06-29 19:44:19	ctrl_and:fr (v3.1.6)	Record mode changed	Start
2017-06-29 19:44:20	ctrl_and:fr (v3.1.6)	Record mode changed	Starting
2017-06-29 19:47:40	ctrl_and:fr (v3.1.6)	Reached peak height	32.6 m
2017-06-29 19:49:43	ctrl_and:fr (v3.1.6)	Log ended	Dist home: 1.4 m; Height: 1.8 m; Rec: 0.0 s
2017-06-29 19:44:06	ctrl_and:fr (v3.1.6)	Photo mode changed	Single

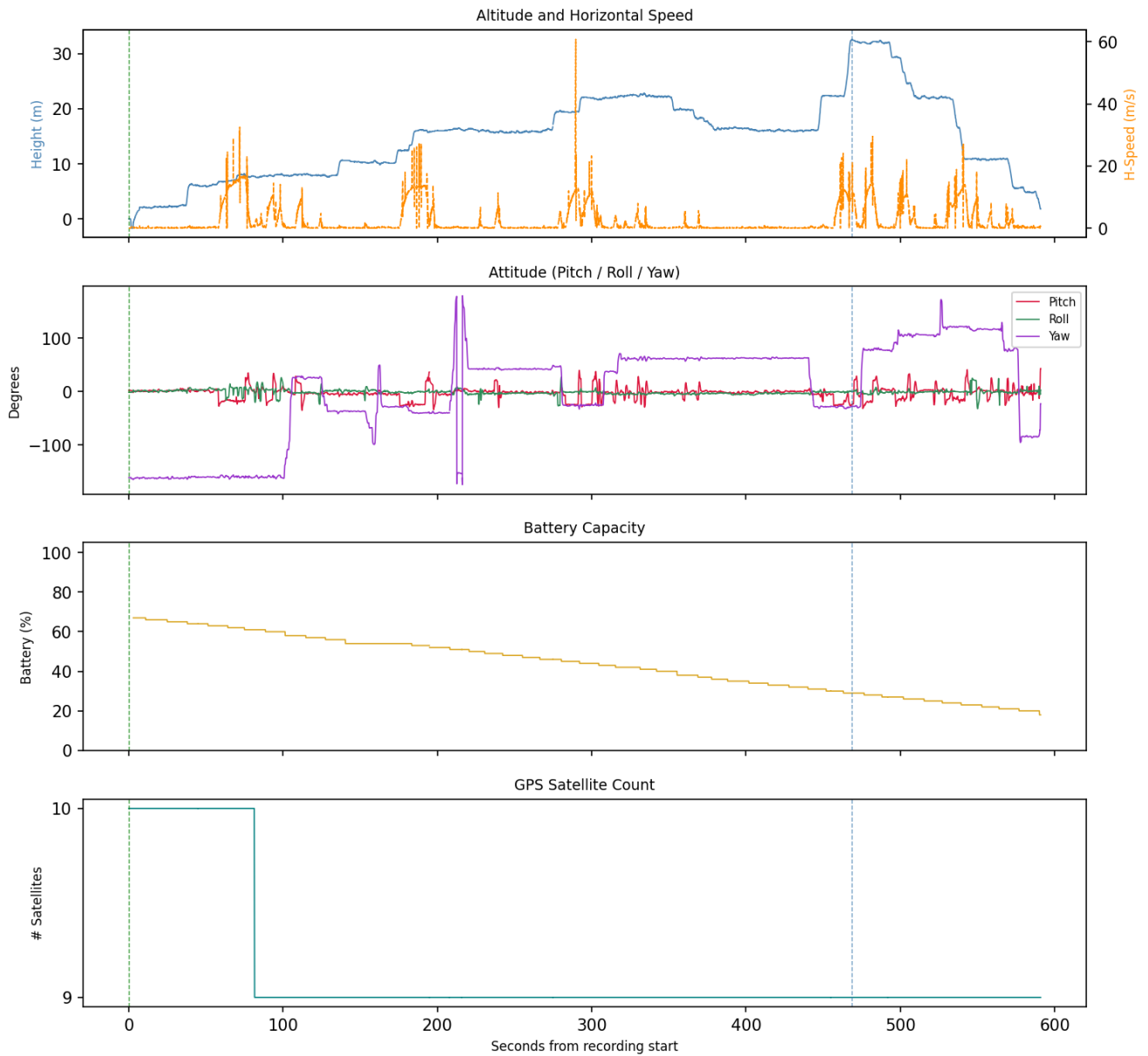
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9596, lon -106.2189)
p5:836a2649180c78233c9be288518bdde5180826defcf8ea0824946cbc6cce8b8c	
drone_sd:videos	GPS within flight bounding box (lat 39.9596, lon -106.2189)
p5:5fb666113f6cb26e6e285a71c38caa890f3f8dfd4491bdd24cec2d8c5a5b106d	

Flight Record - Sensor Telemetry

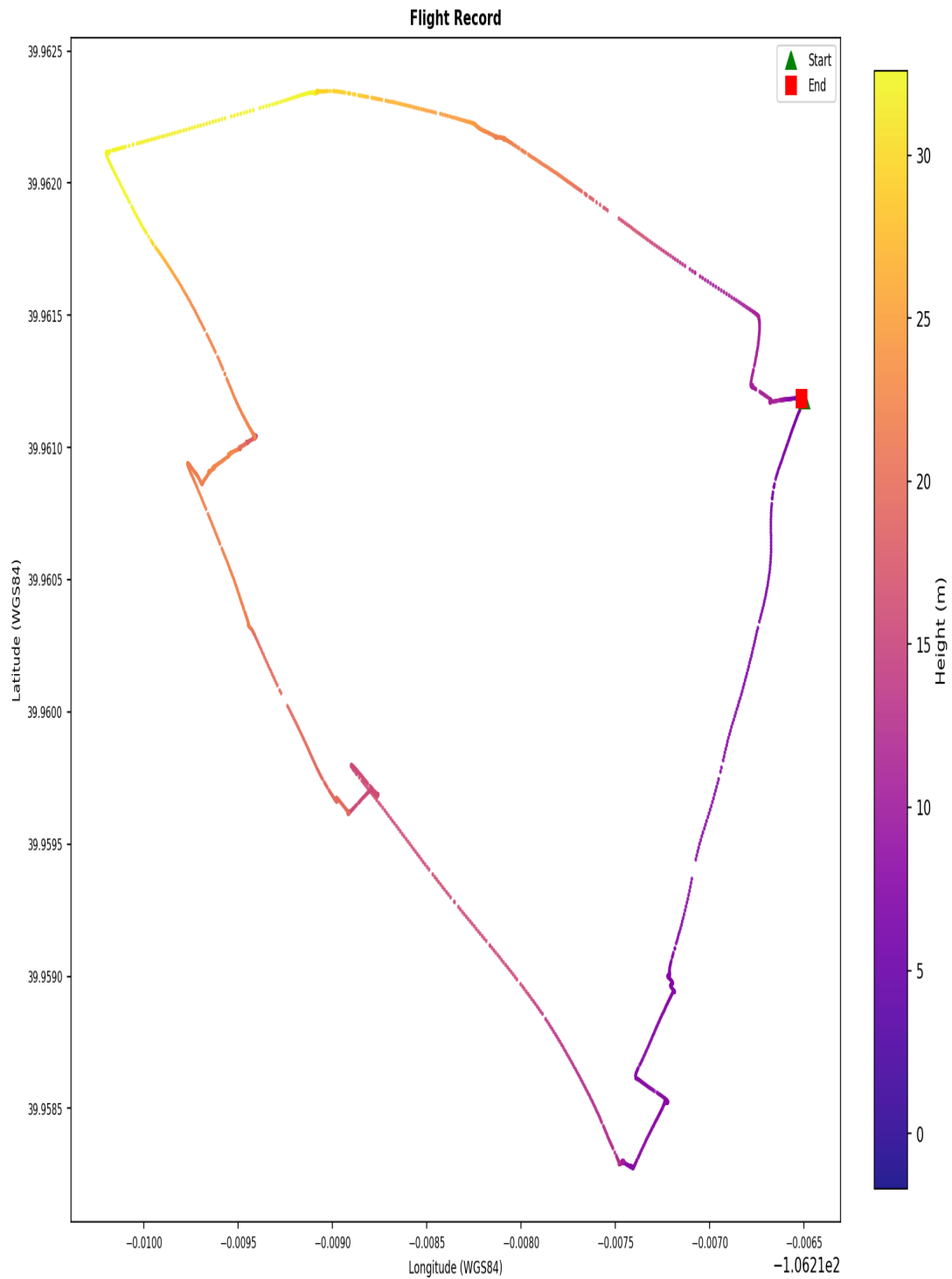
Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_01



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	1	2	0	12	1	0	2
drone_sd	0	0	0	0	0	3	2

Data Quality Notes per Category

Flight Logs (12 files):

- formats: 1 heading_log, 11 unknown

Flight Records (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (3 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 3 file(s) missing (normalisable) EXIF timestamp
- 2 file(s) missing EXIF GPS coordinates

Videos (4 files):

- 3 file(s) contain(s) EXIF GPS coordinates
- 3 file(s) missing (normalisable) EXIF timestamp
- 1 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Flight_logs: log files present in the following format(s): 1 heading_log, 11 unknown.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for df002_flight.zip
2. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for df002_flight.zip
3. [p2 - controller ios]: source evidence not found
4. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller android - images]: no artefacts found
7. [p3 - controller ios]: source evidence not found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Flight Logs

29-06-2017

- format: heading_log | entries: 7

log-2017-06-29.txt

- format: unknown | entries: 0

log-2017-06-29_1.txt

- format: unknown | entries: 0

log-2017-06-29_10.txt

- format: unknown | entries: 0

log-2017-06-29_2.txt

- format: unknown | entries: 0

log-2017-06-29_3.txt

- format: unknown | entries: 0

log-2017-06-29_4.txt

- format: unknown | entries: 0

log-2017-06-29_5.txt

- format: unknown | entries: 0

log-2017-06-29_6.txt

- format: unknown | entries: 0

log-2017-06-29_7.txt

- format: unknown | entries: 0

log-2017-06-29_8.txt

- format: unknown | entries: 0

log-2017-06-29_9.txt

- format: unknown | entries: 0

Flight Records

norm_DJIFlightRecord_2017-06-29_[13-39-52].csv

- altitude_negative: 11 row(s) affected

- contains_no_value: 47 column(s) - APP_SER_WARN.warn, CENTER_BATTERY.isBatteryOnCharge, CENTER_BATTERY.isNeedStudy, CENTER_BATTERY.lastStudyCycle, CENTER_BATTERY.totalStudyCycle (+42 more)
- contains_constant_value: 150 column(s) - APP_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType (+145 more)

Images

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

2017_06_29_13_44_19.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no installed DJI applications found - Device and account information.
2. Note: no controller device name found - Device and account information.
3. Note: no controller device found - Device and account information.
4. Note: no account e-mail found - Device and account information.
5. Note: no last country code found - Device and account information.
6. Note: no aircraft model code found - Device and account information.
7. Note: no controller serial number found - Device and account information.
8. Note: no last known latitude found - Device and account information.
9. Note: no last known longitude found - Device and account information.
10. Note: no account nickname found - Device and account information.
11. Note: no device uuid found - Device and account information.
12. 1 flight record(s) could not be correlated to a drone log (flight_01). Independent corroboration is absent for these flight(s).
13. Photo(s) were recorded during the flight. Copies may reside on both the controller and the drone SD card - both sources should be examined.
14. 2 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

dji.pilot.xml	Account Data	controller_android	extract_logical
SHA-256:	89cccb0960e0cefc8fd2db571f7a14c39894b1e6fa85b8e0adeaab403f81405f		
SHA-1:	55b68ca6eefe4945b71eb141c3e8b218ab26a551		

dji.db	Databases	controller_android	extract_logical
SHA-256:	89ed259e1d088aa18eb6f5c31ba9fa0e7167a2f671ac1d7ac63ee44c5c96683b		
SHA-1:	8d6175c5ac7471576495c685cb2fbceffc261ba9		

logdb.db	Databases	controller_android	extract_logical
SHA-256:	62590edaea94d88ce587e504bab4ab73dd2aed6dc36fec2a628a764e8862ec9d		
SHA-1:	3ba6a20b7beeedec4afb7230facc552925d9670		

29-06-2017	Flight Logs	controller_android	extract_logical
SHA-256:	4d355f2d0ad84271e8269ef4e7926f983e9a8eeec80e19a3330b9a9b63a510cb		
SHA-1:	c21659cdf21d8eae0509b6b026c70fefda0ec5cc		

log-2017-06-29.txt	Flight Logs	controller_android	extract_logical
SHA-256:	e5f92e3b6f5ea6679fa03b6152c78cc59908e245b26ddf501e7b248c849466e2		
SHA-1:	4f5381ecc045cc035374c0b436138cc07d68e6b8		

log-2017-06-29_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	7b99fb94c3cd3b0b7ee7825073d65f00275945ad247f3aec5b2eae44bbf114aa		
SHA-1:	7b22b3fcf75eed121ffd49a9a4f48de85de8e4e2		

log-2017-06-29_10.txt	Flight Logs	controller_android	extract_logical
SHA-256:	cef02a1f4d384486e866ae6a4960b4b2fd439fcde38f76c2c101b76e5963cf52		
SHA-1:	0a39ce791d5789c409d4e58646e0d8f37b6425ce		

log-2017-06-29_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	4b2b961c32aa13abb345d465fe87d79b2a946595f84421ef081392463eecc4ca		
SHA-1:	a5af4e9dd7537bcf593a727dc3861ae6c707c86e		

log-2017-06-29_3.txt	Flight Logs	controller_android	extract_logical
SHA-256:	77842013c23dc8cefce5816510b80f7f0201c008f7cdf2d3662f7c4af5229d20		
SHA-1:	88ab6d69739f48499aa2c3f39ffd18c56b6150fd		

log-2017-06-29_4.txt	Flight Logs	controller_android	extract_logical
SHA-256:	3cd3a853ec0e6b86ca600b2b39ed66f97b78b7a03735fd0c83f21f75942ddf21		
SHA-1:	dae50f7f8ea5dd28466b3b56d0ed18409a5d0b5e		

log-2017-06-29_5.txt	Flight Logs	controller_android	extract_logical
SHA-256:	dd1362b9c0d8e00f1df3d7d0ff9426f38b11a8b4f1922092c529084b108c602b		
SHA-1:	476bf56a891acb37b086486498499392d033462d		

log-2017-06-29_6.txt	Flight Logs	controller_android	extract_logical
SHA-256:	ba220bf87005a1ea99fcb8e397351ef001fb5bee96e402dd1c932234c5e03305		
SHA-1:	7ed69e48eb87c5f22584cc999432704cca1d686d		

log-2017-06-29_7.txt		Flight Logs	controller_android	extract_logical
SHA-256:	6dfe8afd13d0e3a5481b85e688d8595889793fac439dbde294c1e0332b81bafa			
SHA-1:	6d5f2a91983f23cb30486ba7cf5868f38173600b			

log-2017-06-29_8.txt		Flight Logs	controller_android	extract_logical
SHA-256:	2329c42465c85ac3df76c50a73215224cdee1a554f4c0135f6d4dcab2cb80385			
SHA-1:	77a49d165264a851fdf1c954b5290689923fa13c			

log-2017-06-29_9.txt		Flight Logs	controller_android	extract_logical
SHA-256:	d3a35d18cdc85053da4db9c2ec1aedadcd5de230cd54c064d157512c8dd3dfa8			
SHA-1:	7e329189faa9a919015d0d2199438e5714ad133e			

DJIFlightRecord_2017-06-29_[13-39-52].txt		Flight Records	controller_android	extract_logical
SHA-256:	ebaeb3f04f8ed6392ada37f23dd6438c60487e6c1439bc7358db8c58937d73b3			
SHA-1:	fdeee2fcade2395e6773eef9309ce215da0b1ffd			

DJI_0001.THM		Images	drone_sd	extract_physical
SHA-256:	6e8299316aa74b00143c718e087d21ad2b5774c4708060a7c1b3e09ae287c655			
SHA-1:	d4aef438ca2154a7ef557c6224a162bbe35a8547			

DJI_0002.JPG		Images	drone_sd	extract_physical
SHA-256:	8e2a458141656bc8e37c0523c2d42eca57200566a8fdedc32dd87668b93b24c3			
SHA-1:	e8db3bcc57c13fd9445814711fe6a687bbd076f4			

DJI_0003.THM		Images	drone_sd	extract_physical
SHA-256:	4ded0ac63c9f4ed3321dd85ac0cd1fc209cbbc36e84c1ce25c63a6ce88ce6f24			
SHA-1:	e369dc283931f95a9b8371652b78eff5189685f8			

2017_06_29_13_44_19.info		Videos	controller_android	extract_logical
SHA-256:	836a2649180c78233c9be288518bdde5180826defcf8ea0824946cbc6cce8b8c			
SHA-1:	74470b13c6bcc860a312ba7462126e872a642274			

2017_06_29_13_44_19.mp4		Videos	controller_android	extract_logical
SHA-256:	7d5ae874be156848f02c9d1a469f88184cbd8d8bcdef8f8466cb84857c3daa39			
SHA-1:	800f28df92fa375e60dba7ef057d2327fc17c382			

DJI_0001.MOV		Videos	drone_sd	extract_physical
SHA-256:	315aa974792c756b4f17b42006521119a399667af077ac21cfffcc284b3f5d2036			
SHA-1:	481eeec7640bc1d67c7942fa83a326ae4a5cf00			

DJI_0003.MOV		Videos	drone_sd	extract_physical
SHA-256:	5fb666113f6cb26e6e285a71c38caa890f3f8dfd4491bdd24cec2d8c5a5b106d			
SHA-1:	bce75f96a8697ac7607d28644b4f417a18277a48			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	2	0	0
Flight Logs	12	0	0
Flight Records	1	1	1
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	3	3	0
Videos	2	2	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-19 10:52:52 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df002_external.001

2026-06-19 10:52:56 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df002_external.001

2026-06-19 10:53:21 parser_android v- rc=n/a

cmd: df002_flight.zip C:\Users\Floris\Documents\dfdof_output\ft-df002-2017-android-2\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-19 10:53:30 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 9221
out: DJI_0001.MOV

2026-06-19 10:53:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 9222
out: DJI_0002.JPG

2026-06-19 10:53:41 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 9223
out: DJI_0003.MOV

2026-06-19 10:53:46 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 567301
out: DJI_0001.THM

2026-06-19 10:53:46 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df002_external.001 567303
out: DJI_0003.THM

2026-06-19 10:53:48 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTLogToCSVtool.exe DJIFlightRecord_2017-06-29_[13-39-52].txt DJIFlightRecord_2017-06-29_[13-39-52].csv
out: DJIFlightRecord_2017-06-29_[13-39-52].csv

2026-06-19 10:53:50 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_44_19.mp4
out: 2017_06_29_13_44_19_exif.json

2026-06-19 10:53:50 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-19 10:53:50 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG
out: DJI_0002_exif.json

2026-06-19 10:53:51 exiftool v13.57 rc=1

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM
out: DJI_0003_exif.json

2026-06-19 10:53:51 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV
out: DJI_0001_exif.json

2026-06-19 10:53:52 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MOV
out: DJI_0003_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.